VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018

An Internship Report

On

"Full Stack Developer"

01/02/2026 to 01/04/2026

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Engineering

in

Electronics and Communication Engineering

Submitted by

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VEMANA INSTITUTE OF TECHNOLOGY

BENGALURU – 560034

2025-2026

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the internship entitled **"Full Stack Developer"** is a bonafide work carried out by **KAMAL S (1VI22EC066)** in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in **Electronics and Communication Engineering**, Visvesvaraya Technological University, Belagavi, during the academic year 2025-2026.

It is certified that all corrections and suggestions indicated for internal assessment have been duly incorporated in the report. The internship report has been approved as it satisfies the academic requirements prescribed for the said degree.

Internal Supervisor	**External Supervisor**	**HOD**	**Principal**
(Prof. Suma B V)	(Mr. Maneesh Kumar Varma)	(Dr. Suneeta)	(Dr. Vijayasimha Reddy B G)

Name of the Examiners	**Signature with Date**
1. _____	_____
2. _____	_____

ACKNOWLEDGEMENT

I sincerely thank Visvesvaraya Technological University (VTU) for providing me with the opportunity to undertake this internship as a part of the academic curriculum, which has contributed significantly to my learning and overall development.

I express my sincere gratitude to Dr. Vijayasimha Reddy B G, Principal, Vemana Institute of Technology, Bengaluru, for providing the necessary infrastructure, encouragement, and support to successfully carry out the internship work.

I extend my heartfelt thanks to Dr. Suneeta, Head of the Department, Electronics and Communication Engineering, Vemana Institute of Technology, for their constant guidance, motivation, and encouragement throughout the internship duration.

I would also like to thank Internal Supervisor Prof. Suma B V, External Supervisor Mr. Maneesh Kumar Varma, and all Internship Coordinators for their cooperation, coordination, and support during the internship period.

I am thankful to all the teaching and non-teaching staff members of the Department of Electronics and Communication Engineering for their support and encouragement throughout the internship.

I also acknowledge the support of Supram Technologies for providing structured learning resources and an environment that encouraged systematic understanding, practical application, and continuous improvement.

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ABSTRACT

During my internship at Supram Technologies as a Full Stack Developer, I gained exposure to multiple domains including industrial UI design, embedded systems, middleware, and system-level development.

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Chapter 1

INTRODUCTION

This chapter introduces the internship.

1.1 Internship Scope and Objectives

The scope was to gain practical knowledge.

1.2 Relevance of Full Stack Developer to Electronics and Communication Engineering

Full stack development is relevant to ECE.

1.3 Evolution of Practices in the Field

Engineering systems have evolved significantly.

1.4 Current Trends and Technologies

Current trends include embedded AI platforms.

Chapter 2

ORGANISATION PROFILE

Overview of Supram Technologies.

2.1 Introduction

Supram Technologies is a technology company.

2.2 Vision and Mission of the Organization

To be a leader in embedded systems.

2.3 Programs and Services Offered

Internship programs and training workshops.

2.4 Organizational Impact and Reach

Significant contributions to the field.

2.5 Core Values

Innovation, integrity, and excellence.

Chapter 3

WORK DONE / METHODOLOGY

Work completed during the internship.

3.1 Overview of Internship Work

Multiple projects spanning UI design and embedded systems.

3.2 Tools and Technologies Used

React, Node.js, Python, ROS.

3.3 Work Process and Methodology

Agile methodology with weekly sprints.

3.4 Key Project and Case Study

Building an industrial dashboard.

3.5 Challenges Faced

Integrating multiple systems.

3.6 Solutions Implemented

Using middleware for system integration.

Chapter 4

RESULTS AND DISCUSSION

Results and discussion.

4.1 Outcomes of the Work

Several working prototypes.

4.2 Learning Outcomes

Understanding of embedded systems.

4.3 Analysis

Valuable hands-on experience.

Chapter 5

CONCLUSION

Summary of the internship.

5.1 Summary

Comprehensive exposure to full stack development.

5.2 Personal Growth

Improved technical skills.

5.3 Future Scope

Deeper exploration of embedded AI.